

ELA-Guided CNN Screening for JPEG Image Tampering

A concise conceptual pipeline for copy-move and splicing analysis

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ABSTRACT	This paper presents a concise, theory-based workflow that converts JPEG recompression differences into an Error Level Analysis (ELA) map and uses that map as an input to CNN-based image screening. It defines the residual, describes a reproducible data and evaluation protocol, and discusses common confounders. The proposed workflow is not accompanied by new experiments; classifier performance and localization ability remain to be established on source-disjoint data.
KEYWORDS	Error Level Analysis; CNN; JPEG; image forensics; recompression; tampering

1. Introduction

Error Level Analysis (ELA) is a practical way to visualize how image regions change after a controlled JPEG re-save. It is attractive because it produces an image-like residual that can be inspected directly or supplied to a classifier. Prior work has explored both ELA-based CNN classification and recompression-difference learning [1, 2]. This paper develops a concise, theory-grounded pipeline for such screening and identifies the conditions that a proper evaluation should address.

The intended output is a suspicious-image flag with an interpretable residual map. ELA by itself does not prove editing and does not automatically produce a validated tamper mask.

2. Error-level map

Let $I(x,y)$ denote the decoded input pixel and $J_Q(I)(x,y)$ the corresponding pixel after JPEG recompression and decoding at a declared quality setting Q . The absolute difference gives a basic error-level map:

$$E_Q(x,y) = |I(x,y) - J_Q(I)(x,y)|$$

Because the raw differences may be small, a fixed normalization or display gain may be applied for visualization. The same transform must be used for every sample; per-image contrast stretching can conceal meaningful scale differences and create inconsistent model inputs. Areas with unusual residuals are candidates for review, not automatic findings.

3. Proposed ELA-guided CNN pipeline

3.1. Input preparation

Keep the original JPEG unchanged and document the source format. Recompress a decoded copy at a preselected quality setting (or a small, predefined set of settings), decode it, and compute E_Q . Do not repeatedly recompress the source in place. Store the residual together with its settings so the analysis can be reproduced.

3.2. Model task

For image-level screening, use the ELA map as the CNN input and train a binary classifier with authentic and manipulated labels. Copy-move and splicing should be reported separately as well as jointly, since the two operations can produce different traces. A later localization model would require pixel-level masks; a classifier score alone cannot support precise region boundaries. Existing studies show that ELA/CNN and recompression/CNN combinations are established research directions [1, 2], but their reported findings should not be transferred to a new dataset without replication.

3.3. Evaluation design

Construct a dataset with authentic images and controlled edits, keeping all variants from the same source image in one partition. Include edits with rotation, scaling, blending, and varied JPEG save histories. Test on a separate source set and vary the final quality setting. Report precision, recall, F1, and ROC-AUC for image classification; for localization, report overlap metrics against ground-truth masks. Compare an RGB baseline, ELA-only input, and a combined RGB-plus-ELA model to determine whether the residual adds useful information.

4. Failure modes and interpretation

ELA responds to recompression error, not to manipulation as a semantic category. Natural texture, sharp edges, camera-specific processing, resizing, screenshots, and benign edits can create local contrast in the map. Heavy recompression can suppress earlier traces; different quality settings can change which regions appear prominent. Scaling the map too aggressively can amplify noise. A robust report should therefore state Q, scaling, image format, and any preceding transformations, and should avoid a binary forensic conclusion based only on visual brightness.

The proposed evaluation also guards against dataset leakage: random crops from one source image must not be split across training and testing. External validation is necessary before deployment because image sources and distribution pipelines can differ from the training set.

5. Conclusion

ELA can serve as an interpretable compression-aware input for CNN screening when recompression and display settings are controlled. Its value is best assessed against RGB and simple residual baselines on source-disjoint data. Until such validation is completed, ELA maps and classifier outputs should be treated as review aids rather than definitive evidence.

References

- [1] D. Mallick, M. Shaikh, A. Gulhane, and T. Maktum, “Copy Move and Splicing Image Forgery Detection using CNN,” *ITM Web of Conferences*, vol. 44, art. 03052, 2022. doi: [10.1051/itmconf/20224403052](https://doi.org/10.1051/itmconf/20224403052).
- [2] S. S. Ali, I. I. Ganapathi, N.-S. Vu, S. D. Ali, N. Saxena, and N. Werghi, “Image Forgery Detection Using Deep Learning by Recompressing Images,” *Electronics*, vol. 11, no. 3, art. 403, 2022. doi: [10.3390/electronics11030403](https://doi.org/10.3390/electronics11030403).
- [3] N. T. Lam, *Deep Learning Methods for Image Forgery Detection*, undergraduate thesis, Can Tho University, 2026. Chapters 1–2: research overview and theoretical background.

Source note. This concise conceptual paper was prepared from Chapters 1–2 (research overview and theoretical background) of the undergraduate thesis *Deep Learning Methods for Image Forgery Detection* (Nguyen Thuy Lam, 2026). It introduces no new dataset, experiment, or measured performance.